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RESOURCE ANNOUNCEMENT

Genome Sequence Resources for the Wheat Stripe Rust Pathogen (*Puccinia striiformis* f. sp. *tritici*) and the Barley Stripe Rust Pathogen (*Puccinia striiformis* f. sp. *hordei*)

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Abstract

Puccinia striiformis f. sp. *tritici* causes devastating stripe (yellow) rust on wheat and *P. striiformis* f. sp. *hordei* causes stripe rust on barley. Several *P. striiformis* f. sp. *tritici* genomes are available, but no *P. striiformis* f. sp. *hordei* genome is available. More genomes of *P. striiformis* f. sp. *tritici* and *P. striiformis* f. sp. *hordei* are needed to understand the genome evolution and molecular mechanisms of their pathogenicity. We sequenced *P. striiformis* f. sp. *tritici* isolate 93-210 and *P. striiformis* f. sp. *hordei* isolate 93TX-2, using PacBio and Illumina technologies and RNA sequencing. Their genomic sequences were assembled to contigs with high continuity and showed significant structural differences. The circular mitochondria genomes of both were complete. These genomes provide high-quality resources for deciphering the genomic basis of rapid evolution and host adaptation, identifying genes for avirulence and other important traits, and studying host-pathogen interactions.

Genome Announcement

Puccinia striiformis is an economically important fungal pathogen, causing devastating stripe (yellow) rust diseases on cereal crops and grasses (Hovmøller et al. 2011). Within *P. striiformis*, there are several formae speciales, among which are *P. striiformis* f. sp. *tritici* and *P. striiformis* f. sp. *hordei*, highly adapted to wheat and barley, respectively (Chen et al. 1995). Even though few genomes have been available for *P. striiformis* f. sp. *tritici*, they are mainly sequenced using short-read sequencing technologies and are therefore highly fragmented (Cantu et al. 2011, 2013; Cuomo et al. 2017; Kiran et al. 2017; Schwessinger et al. 2018; Zheng et al. 2013). More genomes of *P. striiformis* f. sp. *tritici* with diverse origins and virulence profiles are needed for comparative genomics to precisely pinpoint the pathogenicity determinants. Moreover, genomes of other formae speciales of *P. striiformis* (e.g., *P. striiformis* f. sp. *hordei*) are still lacking, even though they are valuable for studying host adaptation and evolution of formae speciales.

P. striiformis f. sp. *tritici* isolate 93-210 and *P. striiformis* f. sp. *hordei* isolate 93TX-2 were collected in the United States from Montana and Texas, respectively, in 1993. Their formae

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Table 1. Genome statistics of *Puccinia striiformis* f. sp. *tritici* isolate 93-210 and *P. striiformis* f. sp. *hordei* isolate 93TX-2

Statistics	Isolate 93-210	Isolate 93TX-2
Assembly		
Genome size in contigs (bp)	84,627,802	77,368,122
GC content (%)	44.39	44.40
Number of contigs	493	562
Number of contigs >100 kb	243	263
Contig N50	83	110
Contig N50 length (bp)	295,440	218,468
Mean contig size (bp)	171,659	137,666
Annotation		
Protein-coding genes	16,513	15,976
Annotation completeness (%)	95.10	93.00
Transfer RNA genes	648	626
Transposable elements (Mb)	30.46	26.41
Transposable elements (%)	36.03	34.2
Class I (retrotransposons) (Mb)	11.79	9.43
Class II (DNA transposons) (Mb)	10.22	9.14
Others[a] (Mb)	8.45	7.84

[a] Others include sequences of satellites, simple repeats, and low complexity.

speciales status was confirmed by pathogenicity tests of both isolates on both sets of wheat and barley cultivars used to differentiate races of *P. striiformis* f. sp. *tritici* and *P. striiformis* f. sp. *hordei*, respectively. Genomic DNA was extracted as previously described (Xia et al. 2017). A DNA library with fragment sizes of approximately 500 bp was prepared and was sequenced using the Illumina HiSeq PE150 technology. Meanwhile, a PacBio RS II library was constructed for both isolates and was sequenced using P4-C2 chemistry with 16 single-molecule real-time cells, generating 5.55- and 6.27-Gb reads for *P. striiformis* f. sp. *tritici* (93-210) and *P. striiformis* f. sp. *hordei* (93TX-2), respectively, about 70× genome coverage. The Illumina reads were trimmed, corrected, and filtered. In total, 23,750,223 pairs, with a mean read length of 148 bp and comprising 7,030,066,008 bp, passed all tested criteria and were used in subsequent analyses for *P. striiformis* f. sp. *tritici* (93-210), about 87× genome coverage. Similarly, a total of 20,801,841 pairs, with a mean read length of 141 bp and comprising 5,880,525,786 bp, were used in subsequent analyses for *P. striiformis* f. sp. *hordei* (93TX-2), about 70× genome coverage. The high-quality and error-corrected Illumina reads were fed to program Proovread v2.14.0 (Hackl et al. 2014) to correct the potential sequencing errors in the PacBio long reads. Assembler Falcon (with Falcon-unzip) (Chin et al. 2016) was used to de novo assemble the corrected PacBio reads for the nuclear genome, and Canu v1.4 (Koren et al. 2017) was used to assemble the mitochondrial genome. The intermediate assemblies were optimized by program Metassembler v1.5 (Wences and Schatz 2015). The assemblies were finalized and were manually corrected after polishing, using the paired-end Illumina reads and software Pilon (version 1.2.2) (Walker et al. 2014). To improve the accuracy of genome annotation, we sequenced mitochondrial RNA for both isolates. Total RNA was extracted from fresh urediniospores, germinated urediniospores, and infected wheat leaves 2 and 7 days after inoculation. Fresh urediniospores, germinated urediniospores, and infected leaves were ground separately in a mortar in liquid nitrogen. Total RNA was extracted from frozen powder using the Trizol reagent (Thermo Fisher Scientific), following the manufacturer's instruction. RNA quality was checked using the Agilent2100 Bioanalyzer (Agilent Technologies). The RNA samples from different stages of each isolate were mixed into one sample at a 1:1:1:1 ratio for RNA sequencing. The preparation of RNA-Seq library and Illumina HiSeq PE150 sequencing were performed at the Novogen Corporation Inc. Trimmed and corrected RNA-Seq reads were aligned to the reference genome using HISAT2 version 2.1.0 (Kim et al. 2015). An automated genome annotation pipeline MAKER version 3.00 (Holt & Yandell, 2011) was used for gene model calling. The *a. initio* gene prediction was performed using AUGUSTUS, SNAP, and GeneMark programs (Korf 2004; Stanke et al. 2006). Repeats were masked using RepeatModeler (Institute for Systems Biology). We also provided *P. striiformis* f. sp. *tritici* expressed sequence tag and protein sequences as external evidence for gene-model predictions. All predicted gene models and inferred 5′ and 3′ untranslated regions were integrated to produce final gene models. In total, three iterative

runs of MAKER were performed to produce the final gene set for both *P. striiformis* f. sp. *tritici* and *P. striiformis* f. sp. *hordei* assemblies. The annotation completeness was evaluated using BUSCO v2.0.1 (Simão et al. 2015). More detailed procedures on genome assembly and annotation are available upon request, since the space is limited in this announcement.

A total of 84 and 77 Mb were assembled into 493 and 562 contigs for *P. striiformis* f. sp. *tritici* (93-210) and *P. striiformis* f. sp. *hordei* (93TX-2), respectively (Table 1), which were much better in continuity than most previously available *P. striiformis* f. sp. *tritici* genomes (Cantu et al. 2011, 2013; Cuomo et al. 2017; Kiran et al. 2017; Zheng et al. 2013) and in high quality comparable to a recently assembled *P. striiformis* f. sp. *tritici* genome of isolate 104E 137A, also using the PacBio technology (Schwessinger et al. 2018). The assemblies presented here are one haplotype of the dikaryotic genome. We predicted 16,513 and 15,976 protein-coding genes in *P. striiformis* f. sp. *tritici* (93-210) and *P. striiformis* f. sp. *hordei* (93TX-2), respectively (Table 1). Among them, 46.26 and 45.68% were validated by the RNA sequences of these isolates obtained in this study. The annotations of the two genomes have high completeness, at 95.10% for *P. striiformis* f. sp. *tritici* (93-210) and 93.00% for *P. striiformis* f. sp. *hordei* (93TX-2). In addition to nuclear genomes, we also generated the complete mitochondrial (mt) genomes of *P. striiformis* f. sp. *tritici* (93-210) and *P. striiformis* f. sp. *hordei* (93TX-2). The complete mt genomes of both isolates were circular and 101 kb in length with a low GC content (31%).

Since isolates *P. striiformis* f. sp. *tritici* (93-210) and *P. striiformis* f. sp. *hordei* (93TX-2) represent two formae speciales in *P. striiformis* that are highly adapted to different hosts, the genome sequences we present here provide unique resources for studying genomics of host adaptation of devastating plant pathogens in agriculture.

The sequence data have been deposited in the National Center for Biotechnology Information (NCBI) database with BioProject accession PRJNA422914, sample accessions SAMN08200485 for *P. striiformis* f. sp. *tritici* (93-210) and SAMN0800486 for *P. striiformis* f. sp. *hordei* (93TX-2). The assembled and annotated genome sequences have been deposited in NCBI with accessions PKSL00000000 for *P. striiformis* f. sp. *tritici* (93-210) and PKSM00000000 for *P. striiformis* f. sp. *hordei* (93TX-2).

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